

3.4.7 Investigating diversity

Content	Opportunities for skills development
Genetic diversity within, or between species, can be made by comparing: • the frequency of measurable or observable characteristics • the base sequence of DNA • the base sequence of mRNA • the amino acid sequence of the proteins encoded by DNA and mRNA. Students should be able to: • interpret data relating to similarities and differences in the base sequences of DNA and in the amino acid sequences of proteins to suggest relationships between different organisms within a species and between species • appreciate that gene technology has caused a change in the methods of investigating genetic diversity; inferring DNA differences from measurable or observable characteristics has been replaced by direct investigation of DNA sequences. Knowledge of gene technologies will not be tested. Quantitative investigations of variation within a species involve: • collecting data from random samples • calculating a mean value of the collected data and the standard deviation of that mean • interpreting mean values and their standard deviations. Students will not be required to calculate standard deviations in written papers.	AT k Students could: • design appropriate methods to ensure random sampling • carry out random sampling within a single population • use random samples to investigate the effect of position on the growth of leaves. MS 1.2 Students could use standard scientific calculators to calculate the mean values of data they have collected or have been given. MS 1.10 Students could calculate, and interpret the values of, the standard deviations of their mean values.